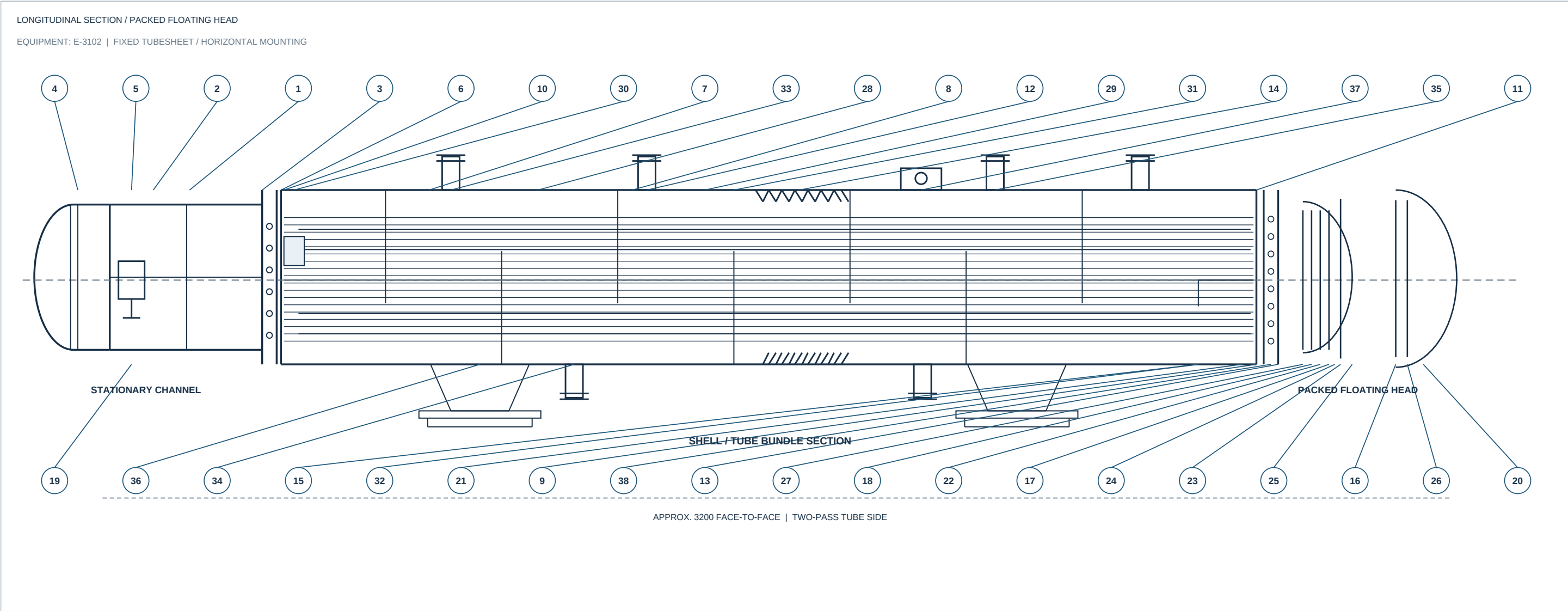


SHELL-AND-TUBE HEAT EXCHANGER

E-3102 | FIXED TUBESHEET / HORIZONTAL MOUNTING

DRAWING: NB-HX-3102-GA-014 REV: C

FICTIONAL PROTOTYPE - NOT FOR FABRICATION



NUMBERED COMPONENTS AND FULL BOM SCHEDULE: SHEET 2 OF 2

GENERAL NOTES

1. Tube layout: 96 tubes on 25.4 mm triangular pitch; two tube-side passes.
2. Shell-side baffles equally spaced; first and last spacing 180 mm nominal.
3. Gaskets are prototype selections only; confirm process compatibility before use.

NORTHBRIDGE THERMAL SYSTEMS ENGINEERING PROTOTYPE FICTIONAL EQUIPMENT AND DATA SHELL-AND-TUBE HEAT EXCHANGER	DRAWING NUMBER NB-HX-3102-GA-014	REVISION C	PROTOTYPE SAMPLE NOT FOR FABRICATION Verify all design data
	SCALE NTS	SHEET 1 OF 2	before engineering use

NUMBERED COMPONENT SCHEDULE AND BILL OF MATERIALS

E-3102 | FIXED TUBESHEET / HORIZONTAL MOUNTING | ITEM NUMBERS MATCH SHEET 1 CALLOUTS

DRAWING: NB-HX-3102-GA-014 REV: C

FICTIONAL PROTOTYPE - NOT FOR FABRICATION

ITEM	DESCRIPTION	MATERIAL	SIZE	QTY	PART NO
1	STATIONARY HEAD, CHANNEL	ASTM A516 GR 70	DN600 / 18 THK	1	CH-3102-01
2	STATIONARY HEAD, BONNET	ASTM A516 GR 70	DN600 / 18 THK	1	BH-3102-01
3	STATIONARY HEAD FLANGE	ASTM A105	DN600 / CLASS 300	2	FL-3102-01
4	CHANNEL COVER, BOLTED	ASTM A516 GR 70	DN600 / 22 THK	1	CV-3102-01
5	STATIONARY HEAD NOZZLE	ASTM A106 GR B	DN100 / SCH 40	2	NZ-3102-01
6	STATIONARY TUBESHEET, DRILLED	ASTM A266 GR 2	OD 600 x 55 THK	1	TS-3102-01
7	HEAT TRANSFER TUBES, SEAMLESS	ASTM A213 TP316L	19.05 OD x 2.11 THK x 3200 LG	96	TB-3102-01
8	SHELL, LONGITUDINAL SEAM	ASTM A516 GR 70	ID 610 x 12 THK	1	SH-3102-01
9	SHELL COVER, FLOATING END	ASTM A516 GR 70	ID 580 x 12 THK	1	FC-3102-01
10	SHELL FLANGE, STATIONARY END	ASTM A105	DN650 / CLASS 300	1	FL-3102-02
11	SHELL FLANGE, REAR HEAD END	ASTM A105	DN650 / CLASS 300	1	FL-3102-03
12	SHELL NOZZLE, FLANGED	ASTM A105	DN150 / CLASS 300	2	NZ-3102-02
13	SHELL COVER FLANGE	ASTM A105	DN580 / CLASS 300	1	FL-3102-04
14	SHELL EXPANSION JOINT	ASTM A240 TP316L	ID 610 / 3 CONVOLUTIONS	1	EJ-3102-01
15	FLOATING TUBESHEET, DRILLED	ASTM A266 GR 2	OD 545 x 48 THK	1	FT-3102-01
16	FLOATING HEAD COVER	ASTM A516 GR 70	DN500 / 20 THK	1	FC-3102-02
17	FLOATING HEAD FLANGE	ASTM A105	DN500 / CLASS 300	1	FL-3102-05
18	FLOATING HEAD BACKING DEVICE	ASTM A516 GR 70	DN500 / 16 THK	1	BD-3102-01
19	FLOATING HEAD NOZZLE	ASTM A106 GR B	DN100 / SCH 40	1	NZ-3102-03
20	FLOATING HEAD COVER, EXTERNAL	ASTM A516 GR 70	OD 680 x 16 THK	1	EC-3102-01
21	FLOATING TUBESHEET SKIRT	ASTM A240 TP316L	OD 548 x 8 THK	1	SK-3102-01
22	FLOATING HEAD PACKING, INNER	GRAPHITE / SS 316	12 x 12 SECTION	4	PK-3102-01
23	PACKING BOX FLANGE	ASTM A105	DN500 / CLASS 300	1	FL-3102-06
24	FLOATING HEAD PACKING, OUTER	GRAPHITE / SS 316	12 x 12 SECTION	4	PK-3102-02
25	PACKING GLAND, FIXED	ASTM A105	DN500 / 16 THK	1	PG-3102-01
26	PACKING GLAND, FOLLOWER	ASTM A105	DN500 / 16 THK	1	PG-3102-02
27	LANTERN RING	ASTM A240 TP316L	DN500 / 18 THK	1	LR-3102-01
28	TIE RODS WITH SPACERS	ASTM A193 B8M	M16 x 980	4	TR-3102-01
29	TRANSVERSE BAFFLES / SUPPORT PLATES	ASTM A240 TP316L	6 THK / 25% CUT	8	BF-3102-01
30	SHELL-SIDE IMPINGEMENT PLATE	ASTM A240 TP316L	8 THK / 260 x 220	1	IP-3102-01
31	LONGITUDINAL BAFFLE	ASTM A240 TP316L	6 THK / 520 x 3200	1	LB-3102-01
32	TUBE-SIDE PASS PARTITION	ASTM A240 TP316L	8 THK / 32 WIDE	1	PP-3102-01
33	VENT CONNECTION, FLANGED	ASTM A105	DN25 / CLASS 300	2	NZ-3102-04
34	DRAIN CONNECTION, FLANGED	ASTM A105	DN40 / CLASS 300	2	NZ-3102-05
35	INSTRUMENT CONNECTION, COUPLING	ASTM A105	DN15 / CLASS 3000	2	IC-3102-01
36	SHELL SUPPORT SADDLE	ASTM A36	610 OD / 10 THK	2	SD-3102-01
37	SHELL LIFTING LUG	ASTM A36	PL 20 THK	4	LG-3102-01
38	FLOATING HEAD WEAR RING	ASTM A240 TP316L	OD 548 x 8 THK	1	WR-3102-01

BILL OF MATERIALS | QUANTITIES ARE PER COMPLETE ASSEMBLY

Drawing schedule is included for extraction testing; verify all quantities and specifications before engineering use.

NORTHBRIDGE THERMAL SYSTEMS ENGINEERING PROTOTYPE FICTIONAL EQUIPMENT AND DATA SHELL-AND-TUBE HEAT EXCHANGER	DRAWING NUMBER NB-HX-3102-GA-014 REVISION C	PROTOTYPE SAMPLE NOT FOR FABRICATION Verify all design data
Two-pass bundle arrangement and exchanger bill of materials	SCALE NTS SHEET 2 OF 2	before engineering use